Lantern

Research you can defend, on infrastructure you control. A working platform — not a concept — that turns public market data and primary filings into auditable research: a calibrated AI-equity risk monitor, grounded filings Q&A, and a research desk that writes and scores its own notes. Built in days, demonstrated live, measurable end to end.

3 modules
live on real data today: monitor, filings Q&A, research desk

100%
abstention recall: no model ever answered an unanswerable probe

77% vs 50%
forecast-band coverage, house model vs Google TimesFM, 252 months

0 cloud calls
needed end to end: answering, forecasting, challenging and judging run locally

What a reviewer sees

- **Every number traces.** Each figure on screen carries its source, pull time and SHA-256; raw responses are kept on disk; notes regenerate bit-for-bit from the stored model run.
- **Honesty is enforced by architecture, not prompts.** A filings answer ships only with quotes that are character-for-character verbatim in the filing, cited to the page — otherwise one fixed abstention sentence. A research-note draft that uses a number absent from the model run is rejected automatically.
- **The models keep their own public scorecard.** An accuracy ledger scores every past forecast against what the market then did; the Q&A system is gated on abstention precision against a human-verified gold set; every claim below is reproducible from the repository.

Measured, not promised

Exhibit	Result	Honest caveat
De-rating probability calibration (2003–2026, 275 months)	Hit rate 0.93 · Brier 0.064 vs 0.064 base	In-sample fit, printed on the page itself; three calibrated drivers, four labelled watch-only
Forecast fan, walk-forward p10–p90 coverage (252 months)	House 77% vs TimesFM 2.5 zero-shot 50% (target 80%)	Challenger runs locally; Apache-licensed weights
Probability challenger (TabPFN tabular foundation model)	Brier 0.079 vs champion 0.077 vs walk-forward base rate 0.040: neither beats it out of sample	Only 7 drawdown months scored, so the model gap is noise; a naive leaky setup scored 0.026 and was caught by test
Filings Q&A abstention gate	Recall 1.00 all models · precision 0.83 on the machine draft	0.90 gate judged on the human-verified set (verification in progress)
On-prem answerer (Qwen2.5-14B behind the same verbatim gate)	0.58 / 0.60 judge-correct (easy/medium) vs cloud 0.83 / 0.80	Local judge (two independent local judges agree within one question); fine-tuning on in-house data measured and documented

All evaluation runs — scoring, judging, fine-tuning, challengers — executed on own hardware with zero external API dependency. Cloud models are an option, never a requirement.

On-premise by design

- **One environment variable** swaps the cloud model for an open-weight model on the bank's own GPU — identical interface, verification, logging; the answer is labelled with its true source and model.
- **Judging is local too:** the evaluation harness scores against a locally served judge, so model-risk evidence is producible inside the perimeter.
- Data, documents, keys and model weights never leave the building; the demo runs from a laptop on live public data each morning.
- **Governance-ready:** model inventory, prompt hashes, provenance, calibration disclosures and challenger comparisons map directly onto the MAS Guidelines on AI Risk Management (inventory expectations ~October 2027).
- **Third-party diligence done:** every component and data source catalogued with its license; the two items needing commercial resolution (PDF library, market-data feed) are identified with costed paths.

Scale path: the whole Singapore market

- Corpus expansion from the eight demo filings to every SGX-listed issuer is **in progress**: issuer universe seeded from SGX's public data, a verified discovery-and-download pipeline running in batches against issuers' own investor-relations sites, with measured coverage per batch.
- Production deployment replaces web harvesting with the bank's licensed SGX/market-data entitlements — the pipeline from ingest onward is identical.
- Index engineering for ~700k passages is scoped with performance gates (retrieval latency budget ~1s).

Why this, why now

- Research output across Singapore banks is thin relative to the data available; no local bank publishes a calibrated valuation-risk metric on AI equities.
- Lantern's answer is measurable outcomes: a number with a public track record, answers with verbatim evidence, and a note a reviewer can regenerate.
- First working version built in under two weeks on public data; every subsequent module landed with tests, independent review and a signed scorecard.
- Next step: a working session with your research and model-risk teams — the platform demos live, including with the network unplugged from every cloud API.